

Negative Externalities: How do Monterey Park's Housing Prices Adjust to Nearby Data Center?

Kimberly Andresen

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Dr. Bruce Brown

Abstract

This paper examines whether data center development generates negative externalities that affect nearby housing prices. Using the transaction data from Virginia, zip code 20105, where the Amazon Web Services (AWS) data center operated in 2024. Using the hedonic pricing model to measure changes within housing values before and after operation. With that, the first regression analysis shows no statistically significant change.

To improve analysis, implementing a difference-in-difference frame, by comparing housing price trends in a treated area (20105) with a nearby control group (20152) which is not nearby to an operating data center. With the estimated treatment effect, it suggested potential decline of 8% in housing prices; it could represent random variation or causal impact.

To assess the implications, applying the estimated effect to the current housing prices in Monterey Park zip code 91755 to which has raised public concern. Using a scenario-based forecasting approach, a modest decline of 3%, a literature declines 6%, and the data pulls 8%.

1. Introduction

With the rapid expansion of data centers across the United States, has come local opposition with the “Not in My Backyard” (NIMBY) mindset. Although data centers are crucial for artificial intelligence growth, the physical presence in residential areas raises concern about environment quality, energy consumption, and impacts on the housing market that surrounds them.

Looking closely into Monterey Park, California, has started the development of a proposed center which sparked significant public backlash. With that, the public organized

protests to express their concerns about air quality and noise from the diesel generators, electricity utility rate increase, and long-term health impacts (Granada, 2026). Residents pushed that the project could create substantial local costs with no benefit to their community.

From an economic standpoint, the concerns raise questions: do data centers generate negative externalities that are dumped onto nearby housing prices? This paper explains this question by applying frameworks and different approaches to evaluate whether this question is true or false.

2. Theory

Housing prices reflect characteristics associated with the property that include structural features, neighborhood attributes, and environmental quality. A negative externality occurs when an economic activity imposes its costs on a third party. Originated from (Pigou, 1920), he explains, “the divergence between private and social net product arises from the fact that one person’s actions may impose costs or benefits on others which are not taken into account.”

With that, large-scale projects, such as data centers, have the ability to generate these externalities through noise, energy consumption, and air pollution. If these do exist, the theory predicts that it’ll reduce the demand for nearby houses thereby decreasing the costs.

To measure these externalities, I used the hedonic pricing framework. Taking inspiration from (Rosen, 1974), housing prices are expressed as a function of characteristics:

$$P = f(z_1, z_2, \dots, z_n)$$

Where P is the housing price, z are the attributes.

3. Literature Review

According to Davis, consumptions of electricity in the United States is forecast to increase 30% between 2008 to 2035 (David, 2011). A big challenge within opening big industrial power plants is the local communities who argue that there is a source of multiple negative externalities that include noise and visual opposition. This paper by David evaluates the claims provided by the citizens using evidence from the housing markets. Using the Hedonic Price Model (Pigou, 1920), David classifies this house as not a “house” but a bundle of characteristics including bedroom and bath, schools, safety, air quality, and proximity. Representing the function as:

$$P(z_1, z_2, \dots, z_n)$$

Where P denotes housing price, and z_i are different characteristics. In his prediction, if a power plant is nearby then houses become less desirable and prices fall. As beta shifts, the housing price P decreases. He denotes the slope as how much households are willing to pay where marginal price = marginal willingness to pay. Learning from his data, Davis only focuses on houses that are 2 miles away from the location. If any farther, his data would be biased due to some plants having more houses in surrounding areas.

Another example of environmental density that provokes great public concerns is hazardous waste sites (Greenstone & Gallagher, 2008). Extending the hedonic housing price framework, Greenstone and Gallagher denote P as marginal implicit price with c_1 being the vector of its characteristics. Equilibrium is generated by consumers and producers assuming that the market is competitive. In contrast to Davis’s paper, Greenstone and Gallagher found that there is no large price rise within the surrounding houses of the plants.

Altogether, both Davis and Greenstone & Gallagher use regressions to estimate the causal effects within each case. Both use the baseline estimate where:

$$Y = B_0 + B_1x + u_i$$

Where Y is the mean house value, B_0 is the house within n miles, x is the price associated with the homes, and u_i is the error. They use the least squares estimation, which is if $u = 0$ or that determinants of price (p) are not correlated with the determinants.

4. Data Analysis

Regression

Using the hedonic regression model to estimate housing price data from zip code 20105 in Virginia around the Amazon Web Serviced (AWS) data center. Started up in 2024, I compared 2023 and 2024 sold housing prices (before and after the data center started running) to see if they changed following the facility’s operations.

Below are the Virginia (20105) housing prices from 2023 to 2024:

Table a (2023):

SOLD DATE	TYPE	ADDRESS	ZIP	PRICE	BEDS	BATHS	SQ FT	LOT SIZE	YR BUILT	\$/SQ FT	HOA/MONTH
November-22-2023	Townhouse	24604 Byrne Meadow Sq	20105	510000	3	2.5	2696		2012	189	318
May-31-2023	Single Family Residential	41147 Hickory Hedge Pl	20105	130000	5	4.5	4792	14810	2019	271	232

Decemb er-12- 2023	Single Family Residen tial	25439 Kinsale Pl	201 05	10300 00	4	3.5	402 6	121 97	200 9	256	99
Novemb er-12- 2024	Townho use	25122 Almandi ne Ter	201 05	73566 0	3	3.5	299 9	261 4	200 9	245	112
May-31- 2023	Townho use	25546 Oak Medley Ter	201 05	70000 0	3	3.5	266 4	304 9	200 9	263	110
June-24- 2024	Townho use	41989 White Mills Ter	201 05	55000 0	3	2.5	221 6		201 8	248	330
March- 31-2025	Single Family Residen tial	41377 Red Spruce Dr	201 05	12500 00	4	4.5	484 3	827 6	202 0	258	169
Septemb er-20- 2024	Townho use	41709 Deer Grass Ter	201 05	92000 0	5	4.5	483 0	479 2	201 2	190	135
May-19- 2023	Townho use	41518 Horse Chestn ut Ter	201 05	52900 0	3	2.5	261 4		201 6	202	390

Table b (2024):

SOLD DATE	PROPER TY TYPE	ADDRESS	ZIP	PRICE	BED S	BATH S	SQF T	LOT SIZ E	YR BUI LT	\$/SQ FT	HOA/ MONT H
Novemb er-12- 2024	Townhou se	25122 Almandine Ter	2010 5	73566 0	3	3.5	299 9	261 4	200 9	245	112
June-24- 2024	Townhou se	41989 White Mills Ter	2010 5	55000 0	3	2.5	221 6		201 8	248	330
Septemb er-20- 2024	Townhou se	41709 Deer Grass Ter	2010 5	92000 0	5	4.5	483 0	479 2	201 2	190	135
March- 31-2025	Single Family	41377 Red Spruce Dr	2010 5	12500 00	4	4.5	484 3	827 6	202 0	258	169

	Resident										
Novemb	Single	25170	2010	12450	5	5	503	740	201	248	121
er-27-	Family	Great	5	00			0	5	0		
2024	Resident	Berkhams									
	ial	ted Dr									
April-12-	Townhou	25411 Elm	2010	62300	3	2.5	182	217	200	341	117
2024	se	Ter	5	0			6	8	6		
Novemb	Townhou	25705	2010	73500	3	2.5	233	261	201	315	132
er-7-	se	Clairmont	5	0			2	4	4		
2025		Manor Sq									
October-	Townhou	26024	2010	84000	4	3.5	320	304	201	262	116
11-2024	se	Huntwick	5	0			1	9	3		
		Glen Sq									
October-	Townhou	25264	2010	66500	3	2.5	257	217	200	258	132
1-2025	se	Curiosity	5	0			4	8	6		
		Sq									
June-30-	Single	25146	2010	97500	4	3.5	410	653	200	238	182
2025	Family	Mineral	5	0			2	4	7		
	Resident	Springs									
	ial	Cir									
Septemb	Condo/C	41900	2010	47000	3	2.5	166		201	283	437
er-27-	o-op	Metamorp	5	0			0		6		
2024		hic Sq									
		#306									

Main key points from these tables include price, square feet, beds, baths, lot size, and sold date. To evaluate the AWS operations on the housing prices, a hedonic regression model was used.

Empirical Model:

$$\ln(\text{price}) = b_0 + b_1(\text{post}) + b_2(\ln(\text{sqft})) + b_3(\text{beds}) + b_4(\text{baths}) + \epsilon$$

Where $\ln(\text{price})$ is the log of home sold price, post is the indicator variable (1) for homes sold after start of 2024, $\ln(\text{sqft})$ is the home size, and b being the percent change in housing prices associated with the operation.

Table 1: Hedonic Regression Results (Zip Code 20105)

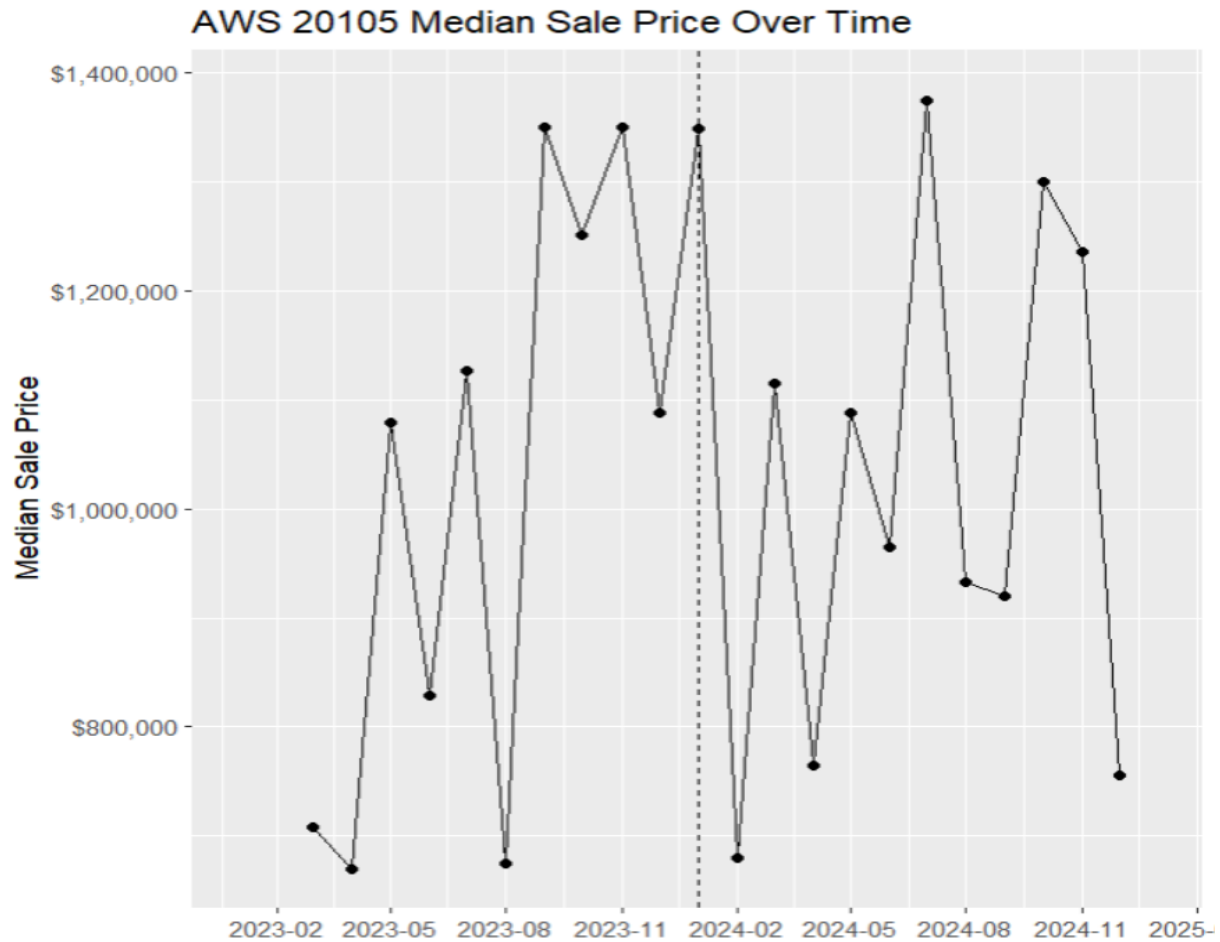
Variable	Coefficient	Std. Error	t-Statistic	p-value
Post	-0.0002	0.1572	-0.0014	0.998
Log(SqFt)	0.392	0.1999	1.96	0.05
Beds	0.0502	0.0917	0.55	0.585
Baths	0.2775	0.0984	2.82	0.005**
Constant	9.0915	1.381	6.58	0.000***

Notes:

- *Observations: 571*
- *Adjusted R² : 0.142*
- **** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$*

The regression results notified that the post estimate is approximately -0.0002 with a p-val of .99885. Indicating that there is no evidence of a change following the data center's operation. Visualizing the median sale price over time, figure 1 plots zip code 20105 from 2023-2024.

Figure 1:



Housing prices fluctuate throughout the sample period, but there is no clear visual break within the graph. In the short run, the housing market doesn't capitalize any negative externalities with the facility.

Since there is no evidence of change, we will form a control and treatment group between zip code 20105 (houses that are around the data center) between zip code 20152 (houses without the data center). The control group is around 10 miles away from the treatment group. Below are the tables for zip code 20152 housing prices from 2023-2024.

Table c (2023):

SOLD DATE	PROPERT Y TYPE	ADDRESS	ZIP	PRICE	BEDS	BATHS	SQFT	LOT SIZE	YR BUILT	\$/SQ FT	HOA/ MONTH
APRIL-5-2024	Townhouse	42793 Shaler St	20152	726000	3	3.5	2448	1742	2001	297	103
JUNE-18-2024	Single Family Residential	25610 South Village Dr	20152	117000	5	4.5	4642	7841	2005	252	122
MAY-10-2024	Single Family Residential	43328 Clarecastle Dr	20152	122500	4	4.5	5341	8276	2005	229	122
FEBRUARY-9-2024	Townhouse	26103 Lands End Dr	20152	625000	3	2.5	2246	1742	2001	278	107
JULY-17-2023	Single Family Residential	26039 Glasgow Dr	20152	112000	5	4.5	4946	8712	1997	226	106
APRIL-12-2024	Single Family Residential	43231 Chase St	20152	931000	4	3.5	3598	8276	1995	259	97
JULY-11-2025	Townhouse	25211 Briargate Ter	20152	560000	3	3.5	2208		2015	254	453
MARCH-6-2024	Condo/Co-op	44075 Peirosa Ter	20152	496650	3	2.5	1598		2014	311	272
SEPTEMBER-27-2023	Single Family Residential	43361 Clarecastle Dr	20152	120000	4	3.5	5381	12632	2005	223	118

Table d (2024):

SOLD DATE	PROPERT Y TYPE	ADDRESS	ZIP	PRICE	BEDS	BATHS	SQFT	LOT SIZE	YR BUILT	\$/SQ FT	HOA/ MONTH
JUNE-18-2024	Single Family Residential	25610 South Village Dr	20152	117000	5	4.5	4642	7841	2005	252	122
MAY-10-2024	Single Family Residential	43328 Clarecastle Dr	20152	122500	4	4.5	5341	8276	2005	229	122

FEBRUAR Y-28- 2025	Single Family Resident ial	43332 Clarecas tle Dr	2015 2	13420 00	4	4.5	550 0	914 8	200 5	244	122
AUGUST- 27-2025	Townhou se	25136 Monteith Ter	2015 2	53500 0	3	3.5	158 2		200 7	338	436
JULY-11- 2025	Townhou se	25211 Briargate Ter	2015 2	56000 0	3	3.5	220 8		201 5	254	453
MARCH- 28-2025	Single Family Resident ial	42521 Mandolin St	2015 2	11511 00	4	3.5	359 5	784 1	200 5	320	98
NOVEMB ER-21- 2024	Single Family Resident ial	43023 Tippman Pl	2015 2	84000 0	3	2.5	274 8	566 3	201 1	306	108
MARCH- 21-2025	Townhou se	42685 Newcom er Ter	2015 2	78100 0	4	3.5	269 7	174 2	200 4	290	108
OCTOBER -3-2025	Townhou se	25359 Wakesto ne Park Ter	2015 2	70050 0	3	3.5	228 8	217 8	201 3	306	130
MAY-19- 2025	Townhou se	42802 Cedar Hedge St	2015 2	67000 0	3	2.5	218 4	174 2	200 1	307	108

Difference-in-difference

Estimating the impact of the data center by creating a difference-in-differences framework compares the change in housing prices over time between treatment and controlled areas.

Difference-in-difference Regression:

$$\ln(\text{Price}) = b_0 + b_1 \text{treatment} + b_2 \text{post} + b_3 \text{treatment} * \text{post} + X + \epsilon$$

Where $\ln(\text{price})$ is the log of housing price, treatment (1) are houses near the data center (20105), treatment (0) are houses away from the data center (20152), post (1) being houses after 2024, post (0) are before 2024 and X are the controlled variables log-sqft, beds, and bath.

Table 2: DiD Regression Results

VARIABLE	COEFFICIENT	STD. ERROR	T-STATISTIC	P-VALUE
TREATMENT	0.094	0.202	0.463	0.643
POST	0.077	0.151	0.51	0.61
TREATMENT × POST	-0.081	0.217	-0.373	0.709
LOG(SQFT)	0.61	0.139	4.387	0.000***
BEDS	0.066	0.062	1.062	0.288
BATHS	0.148	0.068	2.168	0.030**
CONSTANT	7.665	0.938	8.173	0.000***

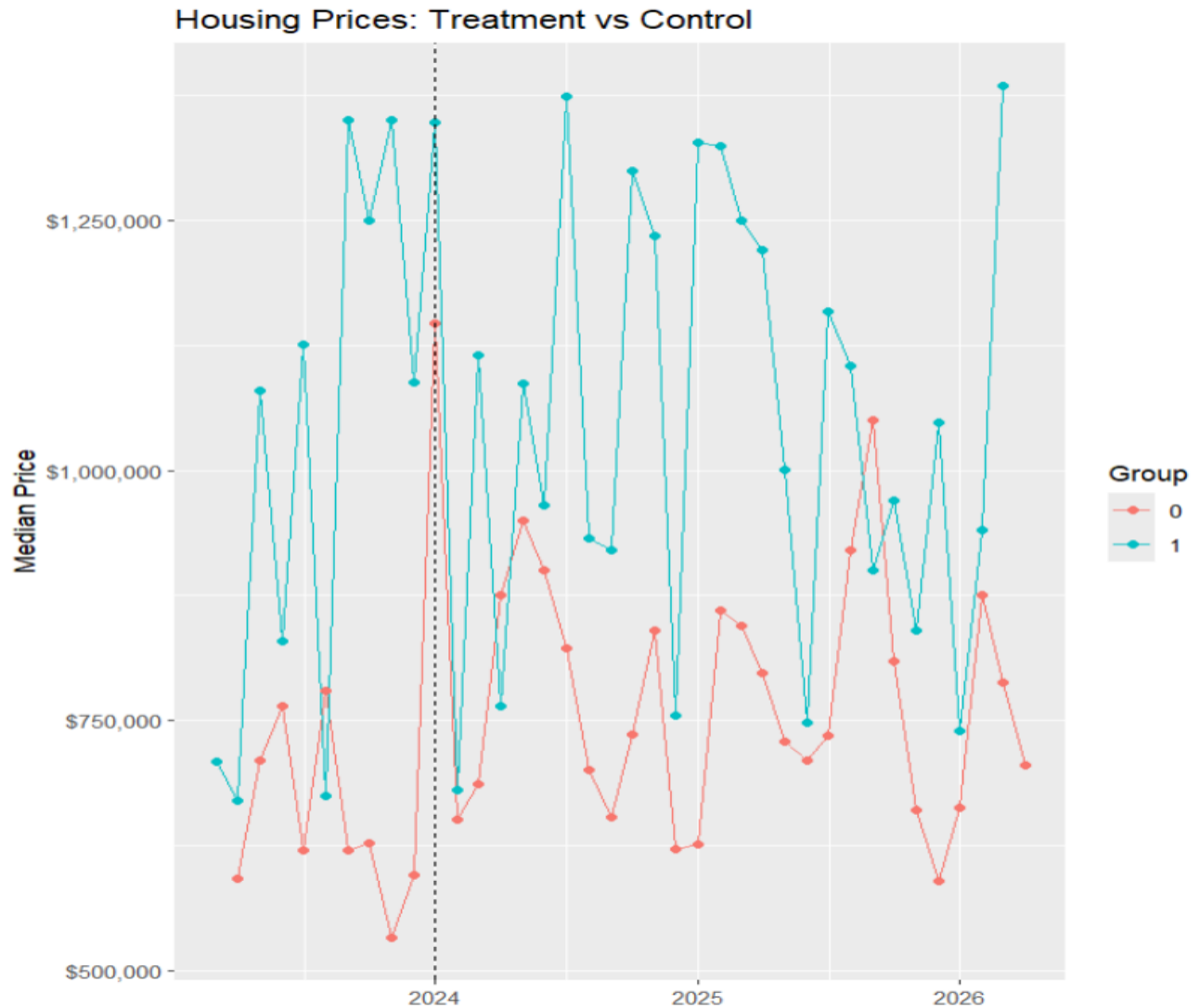
Notes:

- Observations: 1,205
- Adjusted R^2 : 0.113
- *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Regression results indicate that there is a negative effect. The coefficient being an 8% decline in housing prices relative to the control group after the center began operating. Although the large p-value implies that the estimate is not distinguishable from 0. The observed effect reflects the random variation rather than causal impact.

Visualizing the graph, figure 2 plots the difference-in-difference.

Figure 2:



Blue (1) as the treatment group and red (0) as the control group. Where the dashed line is when the data center starts to operate. Both groups have similar patterns of price fluctuations, although the treatment group consistently goes through higher price levels.

There is no clear difference between both groups, in that the two series remain stable with similar levels of volatility. Suggesting that the housing prices in the treated area did not change significantly.

Forecast

If Monterey Park behaves like Virginia, what will happen to the housing prices? Branching off our DiD (difference-in-difference) estimate, where our prices might fall by 8%. We conduct scenario analysis under uncertainty, where our main estimate is our p value - .08. Below are the tables used to estimate the price change.

Table e (2025):

SOLD DATE	PROPERTY TYPE	ADDRESS	ZIP	PRICE	BEDS	BATHS	SQFT	LOT SIZE	YR BUILT	\$/SQFT	HOA/MONTH
OCTOBER-2-2025	Condo/Co-op	227 Florence Ave	91755	1352500	5	5.5	2957	12256	2004	457	30
FEBRUARY-17-2026	Single Family Residential	208 E Newmark Ave	91755	130000	4	3	2528	5107	1999	514	
AUGUST-7-2025	Condo/Co-op	609 E Newmark Ave	91755	104500	4	3	1793	37901	1999	583	150
JANUARY-14-2026	Townhouse	123 N Alhambra Ave Unit D	91755	680000	3	3	1355	17096	1981	502	245
DECEMBER-30-2025	Multi-Family (5+ Unit)	336 S Alhambra	91755	250000	11	5	3398	19007	1951	736	
	Multi-Family (2-4 Unit)	229 S Orange Ave	91755	135000	3	2	1240	6291	1960	1089	
JULY-3-2025	Single Family Residential	501 N Rural Dr	91755	225000	8	8	4197	8033	2002	536	\

Creating a range of plausible outcomes if the estimate is correct; prices will fall by 8%.

We will also include a moderate scenario where a drop off 3% and follow a literature scenario

based on (Davis, 2011) where a 6% price drop is seen. Applying our estimates to find the projected price would be:

Forecast Formula:

$$\text{Projected Price} = \text{Current Price} * e^{b_3}$$

Where b_3 is the log change, and e is the multiplier.

Table 3: Forecasted Housing Prices (Monterey Park)

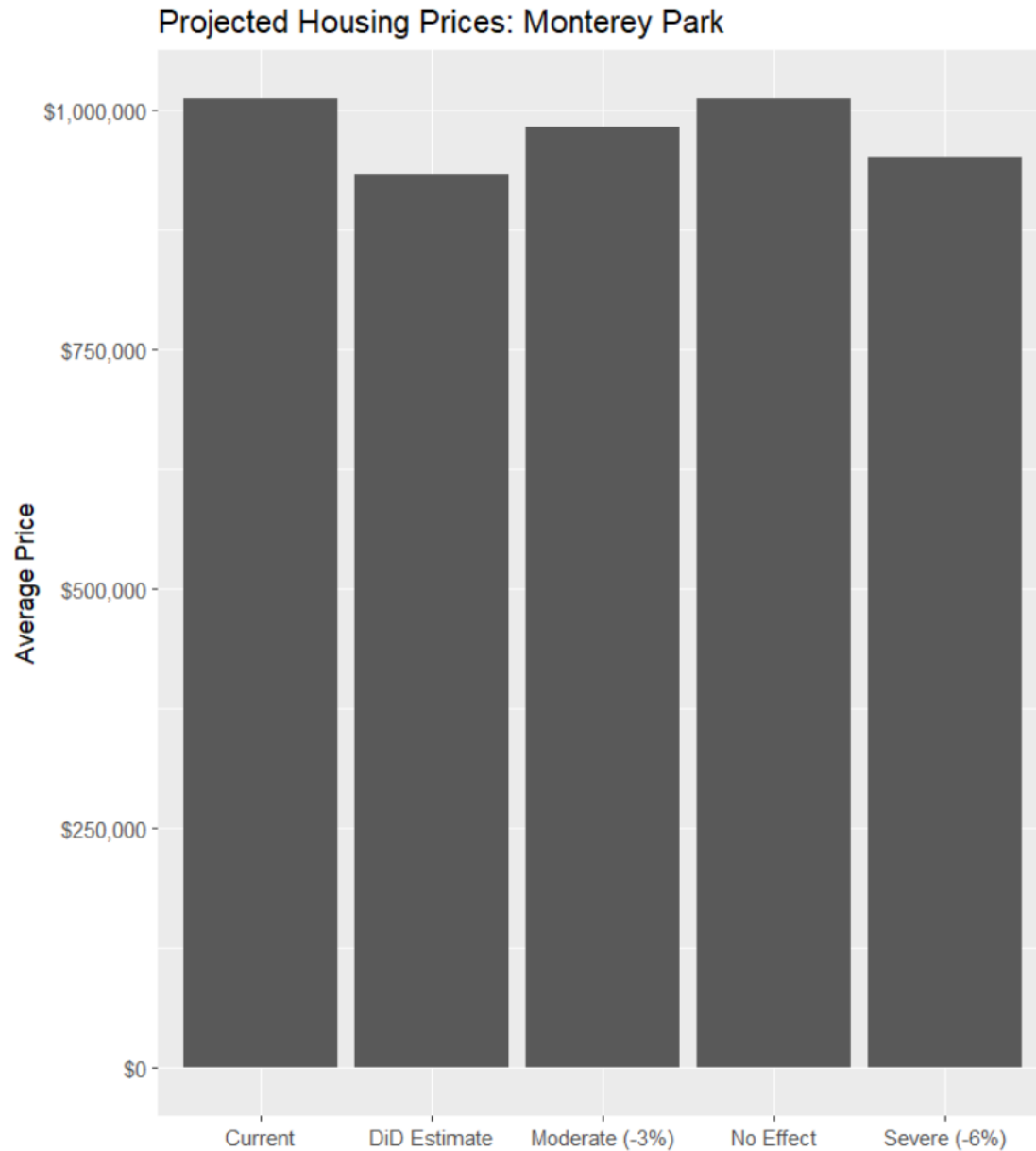
SCENARIO	AVERAGE PRICE (\$)
CURRENT	1,012,745
NO EFFECT	1,012,745
DID ESTIMATE (-8%)	934,171
MODERATE (-3%)	982,363
SEVERE (-6%)	951,980

Notes:

- Moderate and severe scenarios are alternative assumptions
- Prices rounded to the nearest \$

Applying the effect on zip code 91775, the moderate scenario (3%) decline yields an average price of \$982,000, with the severe scenario (6%) based on literature yields and average price of \$952,000. Lastly, applying the DiD estimate (8%) projected an average price of \$934,171. Figure 3 shows the forecast.

Figure 3:



5. Conclusion

Overall, using sold housing data from zip code 20105 Virginia, I estimated the hedonic pricing model to comparing prices before and after the Amazon Web Services (AWS) data center was running in 2024. The results came to no statistically significant change following the operations.

To improve analysis, implementing a difference-in-difference framework by comparing treated area (20105) to the controlled area (20152). Estimates of treatment suggested that there was a potential decline of 8% within the housing prices. Although again, the results also indicated that the result was not statistically significant. This meant that the observed effect may have some random variation rather than causal impact.

Visually, housing prices in the treatment and control group followed similar trends over time. There was no sign of deviation after the data center's operation. According to the data, data centers do not have a high impact on nearby housing prices in the short run.

Even though data implicates little to no change, I still applied the estimated effect to zip code 91775. Using a scenario-based forecast, I generated potential declines that could happen to the housing prices around the data center. A moderate decline of 3%, a literature decline of 6%, and lastly the empirical decline of 8%.

Concluding that data centers may not have large negative externalities to significantly affect housing markets in the short run. The findings do have limitations, such as short time horizons, price volatility, and ZIP level data. To gain a better response or clearer data, future research using longer periods or more ranged controlled data could provide cleaner results. The use of Redfin only provided a 1–5-year span. If I were to obtain past real estate data using Costar, I believe that I could've found something significant up to a 10–20-year span. All in all, community concerns over data center development are understandable, although there is no clear evidence in the short run to support these concerns.

Work Cited

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